

Domain Monitoring and control

code	service	✗ 0						
MC-3	Run time management of HVAC systems	Service group: HVAC interaction control						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Manual setting	0	0	0	0	0	0	0
level 1	Runtime setting of heating and cooling plants following a predefined time schedule	1	1	1	1	0	0	0
level 2	Heating and cooling plant on/off control based on building loads	2	1	2	2	1	0	0
level 3	Heating and cooling plant on/off control based on predictive control or grid signals	3	2	2	3	1	0	0
level 4	0							
Information sources								
Standard?								
code	service	✗ 0						
MC-4	Detecting faults of technical building systems and providing support to the diagnosis of these faults	Service group: Fault detection						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No central indication of detected faults and alarms	0	0	0	0	0	0	0
level 1	With central indication of detected faults and alarms for at least 2 relevant TBS	0	0	0	1	1	1	1
level 2	With central indication of detected faults and alarms for all relevant TBS	0	0	0	2	2	2	2
level 3	With central indication of detected faults and alarms for all relevant TBS, including diagnosing functions	0	0	0	3	3	3	3
level 4	0							

Information sources							
Standard?							

code	service	✖ 0
MC-9	Occupancy detection: connected services	Service group: TBS interaction control

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Occupancy detection for individual functions, e.g. lighting	1	0	1	1	0	1	0
level 2	Centralised occupant detection which feeds in to several TBS such as lighting and heating	1	0	1	1	0	2	0
level 3	0	-	-	-	-	-	-	-
level 4	0							

Information sources							
Standard?							

code	service	✔ 1
MC-13	Central reporting of TBS performance and energy use	Service group: Feedback - Reporting information

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Central or remote reporting of realtime energy use per energy carrier	1	0	0	1	0	1	1
level 2	Central or remote reporting of realtime energy use per energy carrier, combining TBS of at least 2 domains in one interface	1	0	0	2	0	2	2
level 3	Central or remote reporting of realtime energy use per energy carrier, combining TBS of all main domains in one interface	1	0	0	3	0	3	3

level 4	0							
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Information sources							
	Standard?						

code	service	✓ 1	
MC-25	Smart Grid Integration		Service group: Smart Grid Integration

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None - No harmonization between grid and TBS; building is operated independently from the grid load	0	0	0	0	0	0	0
level 1	Demand side management possible for (some) individual TBS, but not coordinated over various domains	0	2	0	0	0	0	0
level 2	Coordinated demand side management of multiple TBS	1	3	0	1	0	0	0
level 3	0							
level 4	0							

Information sources							
	Standard?						

code	service	✗ 0	
MC-28	Reporting information regarding demand side management performance and operation		Service group: Feedback - Reporting information

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Reporting information on current DSM status, including managed energy flows	0	1	0	0	0	0	2
level 2	Reporting information on current historical and predicted DSM status, including managed energy flows	0	2	0	0	0	1	3
level 3	0							
level 4	0							

Information sources							
	Standard?						

code	service
MC-29	Override of DSM control

✗ 0

Service group: Override control

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No DSM control	0	0	0	0	0	0	0
level 1	DSM control without the possibility to override this control by the building user (occupant or facility manager)	0	3	-2	0	0	-1	-2
level 2	Manual override and reactivation of DSM control by the building user	0	1	0	1	0	0	0
level 3	Scheduled override of DSM control (and reactivation) by the building user	0	1	0	2	0	1	0
level 4	Scheduled override of DSM control and reactivation with optimised control	0	2	0	3	0	1	0

Information sources							
	Standard?						

code	service
MC-30	Single platform that allows automated control & coordination between TBS + optimization of energy flow based on occupancy, weather and grid signals

✓ 1

Service group: ation between TBS + optimization of ei

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Single platform that allows manual control of multiple TBS	0	0	0	1	0	1	0
level 2	Single platform that allows automated control & coordination between TBS	1	0	0	2	0	1	0
level 3	Single platform that allows automated control & coordination between TBS + optimization of energy flow based on occupancy, weather and grid signals	2	0	0	3	0	1	0
level 4	0							

Information sources							
	Standard?						